

SEGREGATED APPROACH FOR THE MODELING OF CONTINUOUS HEAT TREATMENT FURNACES

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ABSTRACT. Accurate modeling of the transient material temperature distribution inside the continuous heat treatment furnaces are very critical in terms of adjusting the process parameters of the furnace. In this study the transient heating process inside a continuous glass annealing furnace is simulated using a segregated approach. In this regard, the convection inside the furnace is solved with a separate steady solver in the Eulerian frame of reference and the transient conduction inside the moving glasses is solved in the Lagrangian frame of reference. Radiation heat transfer between the glasses and the furnace is modeled using a graphics processor accelerated Monte Carlo ray tracing based surface-to-surface radiation model. The glass and furnace surfaces are taken as black in the considered scenario. The segregated approach is evaluated to be suitable as a modeling approach in terms of the convergence rate. Overall, solving for the present continuous annealing furnace with complex glass geometries took around 2 hours of computational time. Finally, the observed non-uniformity in the glass temperature distribution at the furnace outlet is quantified and the possible causes of this non-uniformity are discussed.

1. INTRODUCTION

Continuous heat treatment processes are very commonly used in glass, metal and chemical industries. It is vital to precisely know and control the temperatures of the objects inside the heat treatment furnaces for the products to obtain the desired mechanical properties. However, often the actual temperatures of the objects inside the furnace is not measured but instead the air temperature in the furnace is monitored. Nevertheless, especially in high temperature furnaces there might be a considerable difference between the solid and the air temperatures due to the effect of the thermal radiation.

Simulating the heat treatment process inside continuous furnaces is challenging. It requires solving for all the heat transfer mechanisms in computational domains as large as 40 m or more with mesh resolutions down to milimeter range to resolve the large temperature gradients inside the material. This disparity of the length scales in the problem makes the problem computationally very intensive. Moreover, as opposed to the batch furnaces, the movement of the objects inside the continuous heat treatment furnaces increases the complexity of the problem.

Annealing is a type of the heat treatment process, where the material is heated to a certain temperature and let to cool down in a very carefully adjusted rate. It has a very direct influence on the mechanical properties of the final product. Modelling the temperature transients inside the material and adjusting the process parameters according to the requirements is very crucial for the process. Prieto et al. [1] developed a stepwise thermal model for an annealing furnace line. In their model the furnace was

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divided into several enclosures in which overall heat balances were conducted. Moreover, the mechanisms of both heat convection and radiation were considered in the analysis of steel strips. Depree et al. [2] developed a full three-dimensional model for a continuous steel annealing furnace and came up with a 1D/2D simplified and computationally fast model of the same furnace to be used for studying alternative operating conditions of the furnace. The modelling of a direct-fired continuous annealing furnace, considering the chemical reactions and the flue-gas composition for a steel strip is carried out by Strommer et al. [3] Similarly, Zareba et al. [4] presented a mathematical model of stainless-steel annealing furnace considering radiative and convective mode of heat transfer for the calculation of strip, wall and gas temperatures. Jaklic et al. [5] considered a walking beam type steel reheating furnace where they have solved for the radiative heat transfer using a view factor model and a finite difference method for the transient conduction inside the steel. Han and Chang [6] have simulated a steel reheating furnace where they have solved for the combustion in addition to the transient heat conduction in the steel slab. They considered the movement of the steel slabs in a discontinuous manner where they changed the position of the slab instantly after a certain heating time. Singh and Salukdar [7] have considered the movement of the steel slabs in a similar discontinuous manner. Even though they have ignored solving for the combustion zone the simulations took around 100 h. Margado et al. [8] simulated a walking beam type steel reheating furnace where they have solved both for the combustion zone inside the furnace with the conduction inside the metals using a transient solver. They compared this high fidelity approach with a more simplistic approach where the zone temperatures of the furnace were considered as uniform. Antonio et al. [9] modeled the transient heating process of a moving steel block in a vertical experimental furnace. For modeling of the movement of the steel block they have considered a low fidelity field assignment method similar to the ones used in steel reheating furnaces [6,7,8] and a high fidelity dynamic mesh method. They have concluded that the dynamic mesh method is more suitable for the simulation of continuous processes.

In this present numerical study, a segregated numerical approach has been used for the modelling of the heating section of a continuous glassware annealing furnace operating at realistic conditions. For this study wineglasses are selected as the glassware to be annealed. This study is first in the sense that it deals with heat treatment process of geometrically complex objects. Regarding the segregated approach, the convection inside the furnace is first solved in the Eulerian frame of reference without considering the movement of the glasses inside the furnace and next, the transient conduction inside the glass is solved in the Lagrangian frame of reference considering the movement of the glasses and imposing the convective and radiative heat fluxes on the glass outer boundary as heat flux boundary conditions. For calculating the radiative heat fluxes on the moving glass geometries, an in-house developed graphic processor accelerated Monte Carlo ray tracing (MCRT) based surface-to-surface radiation model has been used. In the results section, the convergence behavior of this iterative segregated approach is first discussed and next, the effectiveness's of the convective and radiative heating mechanisms are evaluated. Finally, the causes of the observed non-uniformity in the temperature distribution inside the glasses at the outlet of the heating section are discussed.

2. NUMERICAL METHOD

Glass annealing process inside continuous annealing furnaces is a transient process that involves all the three heat transfer mechanisms. While the glasses are moving inside the furnace heat is transferred to the glass surface via thermal radiation and convection. The incoming heat flux on the glass surface is then conducted into the interior of the glass via conduction. The continuous annealing furnace model considered in this study is shown in Figure 1. The computational domain covers only the heating section of the furnace. As shown, 16 rows of glasses are present inside the furnace and each glass row is composed of five glasses. The furnace walls are set to a temperature of 650°C and hot air at 650°C is blown from the bottom of the furnace. The glasses and furnace walls are considered

black and the atmosphere inside the furnace is considered as non-participating. Apart from these boundary conditions about the furnace itself, the temperature of the first row of glasses entering into the furnace is set to a uniform value of 300°C . The operating parameters of the furnace is given in Table 1.

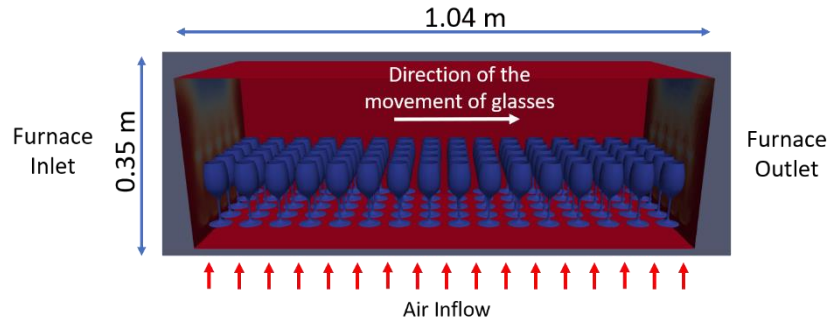


Figure 1. Continuous Furnace Model

Table 1. Operating Conditions of the Furnace and Radiative Properties of the Furnace and Glassware

Temperature of the Furnace walls	650°C
Air Inflow Speed	2 m/s
Air Inflow Temperature	650°C
Initial Temperature of the Glasses at the Furnace Inlet	300°C
Conveyer speed	0.65 cm/s

To ensure the same mesh resolution in the air surrounding the glass and inside the glass across all the 16 rows of glasses, both the solid glass mesh and the air mesh have been generated for one row of glasses only. The generated mesh is then replicated for all the glass rows inside the furnace. This ensured that there is no difference in mesh properties between the different sections of the furnace. To capture the convective heat transfer coefficient accurately, a prism layer mesh has been used to resolve the thermal and viscous boundary layers around the glass. (Figure 2) Specifications of the mesh is given in Table 2.

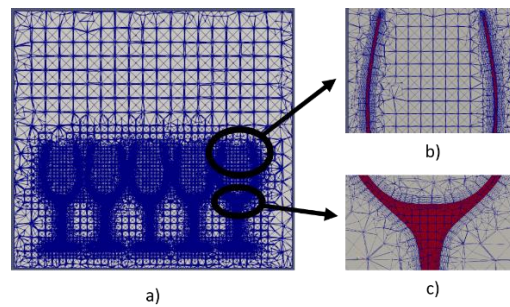


Figure 2. a) Mesh cut through the glasses, b) Mesh in the vicinity of the glass tip c) Mesh in the vicinity above the glass leg

Table 2. Number of cells used in the mesh

	Air Mesh	Solid Mesh
Number of Cells in one Glass	-	0.066M
Number of Cells in a single row (5 Glasses)	0.85M	0.33M
Total Number of Cells (80 Glasses)	13.5M	-

In this study, the transient annealing process is solved in a segregated approach. As shown in the flow chart in Figure 3, first the convection inside the furnace is solved in Eulerian frame of reference by

assuming a uniform temperature distribution among all the glasses inside furnace and next the transient heat conduction inside the glasses is calculated in Lagrangian frame of reference by imposing the calculated convective and radiative heat fluxes on the glass surfaces as boundary conditions. Next, the calculated glass surface temperatures are used again in the convection simulation to re-calculate the convective heat fluxes on the glasses to be used in the transient heat conduction calculation again. This process goes on until the change in the glass surface temperatures are below the predefined tolerance.

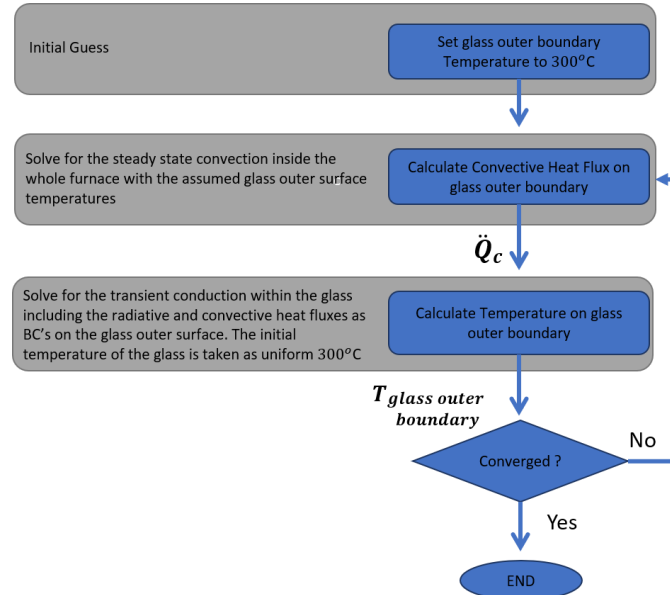


Figure 3. Flowchart for the segregated solution approach

The convection simulation has been carried out with the steady “buoyantSimpleFoam” solver within the OpenFOAM framework. The steady state solution approach for solving the convection inside real continuous furnaces is valid to a large extent considering the presence of a furnace control system which maintains the air temperature at each section of the furnace at the desired level. After solving for the convection inside the furnace, the transient heat conduction inside the moving solid glasses is next studied in the Lagrangian frame of reference. This part of the solution approach deals with a combined transient conduction and thermal radiation problem between opaque surfaces. The challenge in this scenario is the temporal change in the radiative fluxes due to the movement of the glasses inside the furnace. As shown in Figure 4, the first three rows of glasses next to the furnace inlet are positioned into an empty furnace at the start of the simulation and the glasses are displaced by one row at a time towards the outlet of the furnace until the glasses reach the outlet of the furnace. The temperature of the glasses at the entry on the furnace is taken as uniform 300°C as initial condition. Among those three rows of glasses the temperature distribution is calculated only for the glass row at the center of the other two rows of glasses. The additional two rows of glasses surrounding the center row are only due to incorporating the thermal radiation transfer between the glass rows correctly.

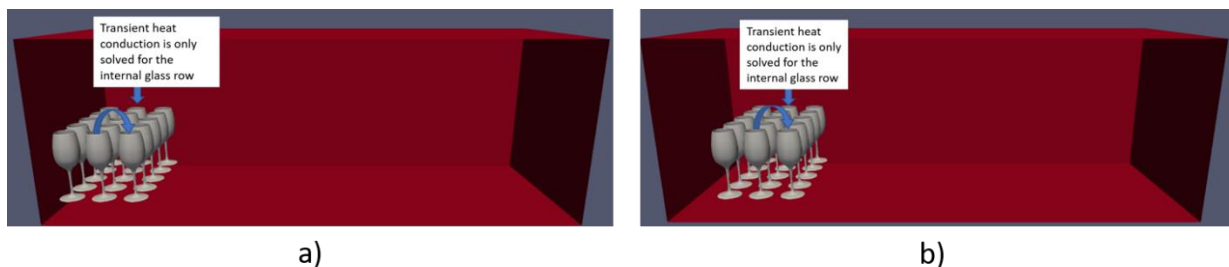


Figure 4. a) Initial position of the glasses at the first time level b) The position of the glasses at the next time level

The radiative heat fluxes on the glass surface is calculated using the in-house developed surface-to-surface radiation model. The calculation considers radiative transfer between the walls of the furnace and the glasses and between the glasses in the same row and in the neighboring rows. A backward Monte Carlo ray tracing method is used for the calculation of the radiative fluxes. 2.5 billion rays are emitted from the surface elements at the center row of glasses for each position of the glasses inside the furnace. A total of 40 billion rays are traced during the movement of the glasses from the furnace inlet to the furnace outlet. The computational cost of the surface-to-surface radiation model is alleviated by the use of a graphic processor. The calculated radiative fluxes calculated on the graphic processor are transferred to the CPU where the transient heat conduction is solved in OpenFOAM with the radiative and convective heat fluxes imposed as heat flux boundary conditions on the glass outer surface.

3. RESULTS

The convergence behavior of the segregated approach on solving the transient heat transfer process in the specified continuous furnace model is evaluated based on the change in the temperature distribution on the glass outer surface with increasing number of iterations. In this regard, a single temperature monitoring point is selected on the outer boundary of each of the sixteen glass rows. Figure 5 shows the temperature levels at each of the monitoring points. Initially, before solving for the conduction inside the glass the temperatures in all the sixteen rows are set to a uniform temperature of 300°C. Next, the convective heat fluxes on the glasses are calculated by solving for the convection inside the furnace with the glass surface temperature all set to the defined initial guess. As the second step, the transient conduction inside the glass is solved for only one row of glasses imposing both the previously calculated convective heat fluxes and the radiative fluxes on the glass outer surface. Finally, the temperatures at the monitoring points at the outer boundary of the glasses are extracted. As shown in Figure 5, the temperature levels at the monitoring points show a major rise at the end of the 1st iteration. At the end of the second iteration a slight decrease in the temperature levels is observed and the third iteration does not lead to a considerable change in the temperature levels at the selected monitoring points.

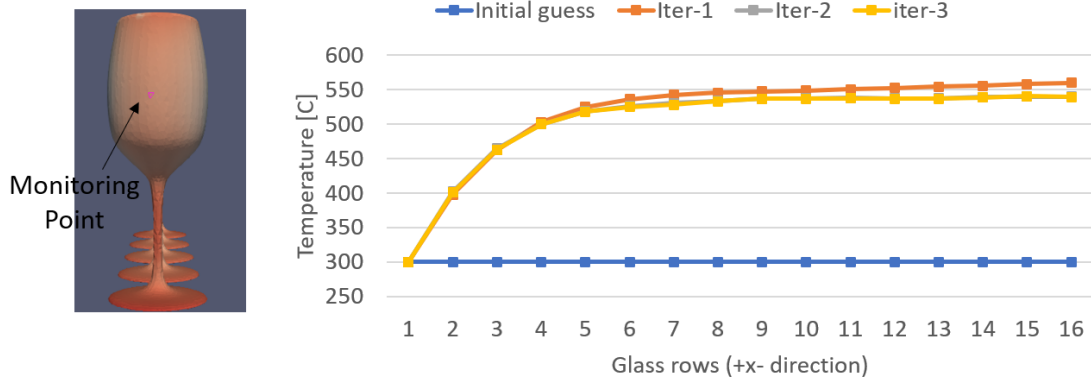


Figure 5. Temperature distribution at the indicated monitoring point across the rows of the glasses at different iteration levels

Similarly, the convergence behavior of the method is also discussed on the detailed temperature distribution on the glass outer boundaries inside the furnace as shown in Figure 6. Also here a similar behavior is observed and the temperature field seems to be converged to a large extent after two iterations. But, the temperature distribution at the lower half of the glasses still shows a considerable change in temperature indicating the necessity of another iteration.

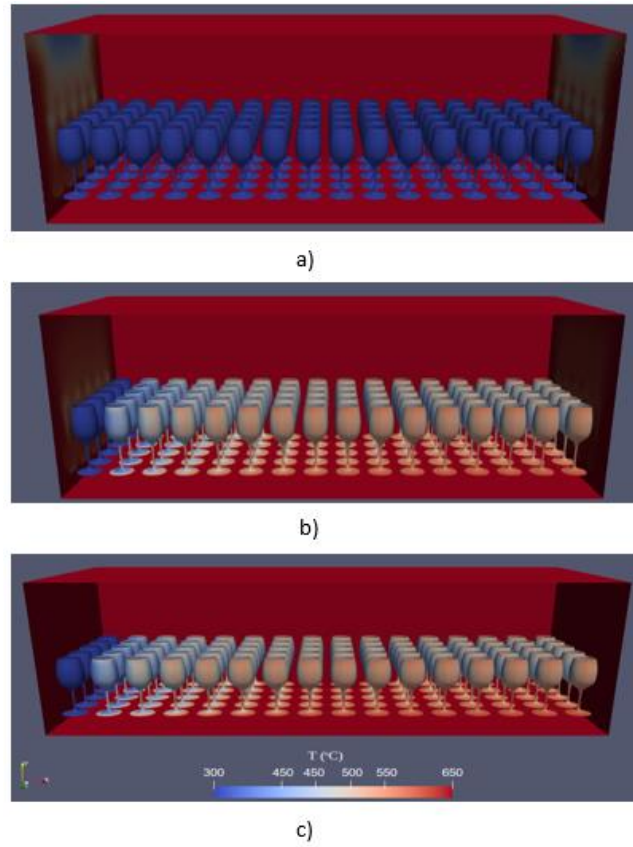


Figure 6. Temperature distribution at the glass outer surface a) initial guess b) end of first iteration c) end of second iteration

The relatively fast convergence rate of the segregated approach applied in this study is possibly due to the dominance of the radiative heat transfer in the heat treatment process. Figure 7 shows the convective and radiative heat fluxes calculated on the glass outer boundary. As shown, the radiative heat flux is about four times as big as the convective heat flux on the glass surface. The iterations are needed only to correct for the convective heat fluxes due to the change in the glass outer temperatures. However, since the convective heat fluxes are much lower than the radiative fluxes in magnitude, the corrections are rather small in magnitude. However, one needs to point out that at the regions where the effectiveness of the radiative heating is low, the method might need more iterations to converge. This might be the case for the upper half of the glasses that are away from the side walls of the furnace.

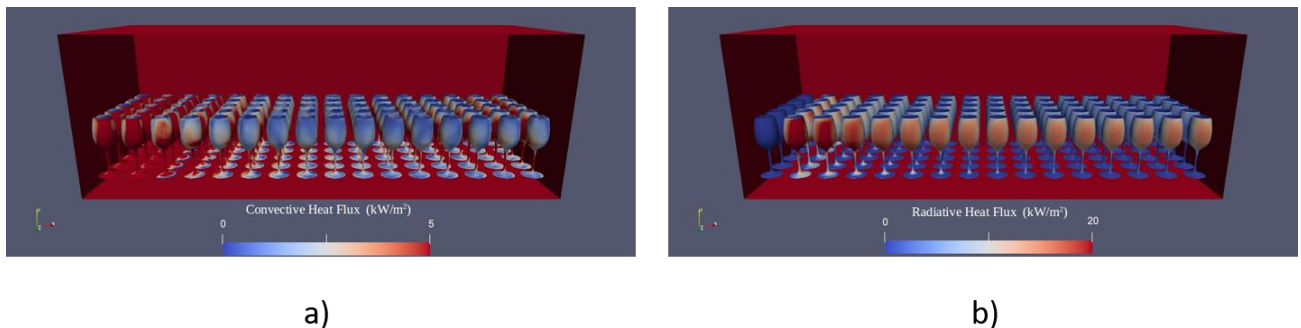


Figure 7. a) Convective Heat Fluxes b) Radiative Heat Fluxes

The internal temperature distribution of the glasses the furnace outlet is shown in Figure 8a. As shown, the temperature levels at the lower half of the glasses is considerably higher compared to the upper half. The temperature levels on the lower half are around 600°C . On the upper half, the glasses next to the furnace walls are heated most. However, the heating is considerably non-uniform. The

surfaces of the glasses facing the furnace side walls reach to around 550°C but the opposite surfaces show a very mild heating. Similarly, the upper half of the interior glasses show relatively low temperature rise however, their temperature distribution is more uniform compared to the glasses on the sides. The non-uniformity of the temperature distribution is also shown on the indicated temperature monitoring points A and B across the rows of the glasses inside the furnace shown in Figure 8b.

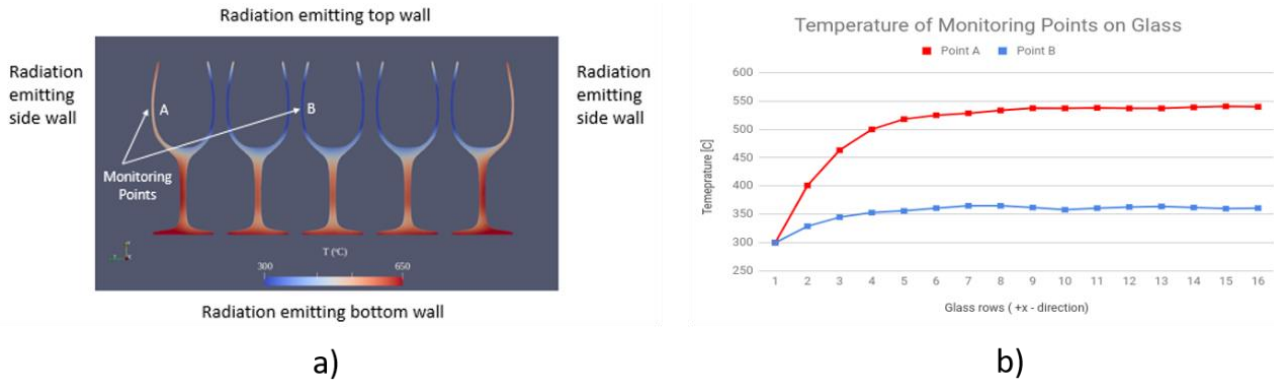


Figure 8. a) Internal temperature distribution at the outlet of the furnace. b) Temperature at the indicated monitoring points A and B across the glass rows inside the furnace

The difference in the dimensions of the glass between the lower and upper halves reflects also on the temperature distribution inside the glass. In the lower half the legs of the glasses are further apart from each other compared to the upper half where the glasses are in a very tight arrangement. Due to the rather small spacing between the glasses, the thermal radiation emitted by the side walls of the furnace is almost entirely blocked by the glasses next to the side walls and the internal glasses receive a very low radiative heat flux. On the contrary, the legs of the glasses are rather distant from each other. So, the legs do not block each other's thermal radiation as much as the upper halves of the glasses do and therefore the heating at the lower part of the glasses are much more effective.

At last, the computational cost of simulating the heating process is evaluated. Figure 9 shows the computational cost of solving for the different modes of heat transfer mechanisms present in the furnace. The steady convection has been solved inside the whole furnace consisting of a mesh size of 13.5 million cells. 16 CPU cores have been utilized for this calculation. The transient conduction inside the solid is solved for a single glass row only consisting of an overall mesh size of 0.33 million cells and utilizing a single CPU core. Simultaneously, the radiation between the surfaces are also solved using the in-house developed radiation model running on a single Nvidia Titan X graphic card. Overall computation time is measured to be around half an hour for a single iteration. In a scenario where 4 iterations would be enough for convergence overall run time would be around 2 hours.

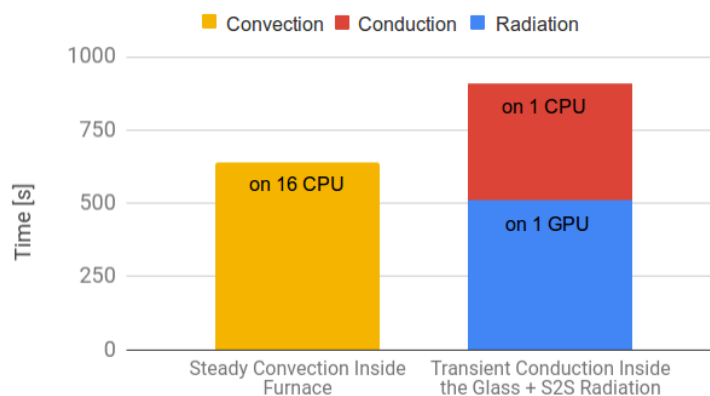


Figure 9. Computational Time for the Continuous Furnace Simulation

4. CONCLUSION

In present study, a segregated solution approach has been used for the modelling of the heating section of a continuous glass annealing furnace. This numerical work is first of its kind regarding the geometrical complexity of the objects inside the furnace. It is shown that the approach converges in a few iterations leading to overall computational time of around 2-3 hours for the modeling of the process. The fast convergence rate of the method is attributed to the dominance of the radiation heat transfer in the furnace model studied. A quite non-uniform temperature distribution is observed inside the glass at the furnace outlet. Radiative heating is found to be much more effective in heating the lower part of the glasses in comparison to the upper halves of the glasses. The discrepancy in the heating behaviors is thought of the difference in the spacings between the glasses in the lower and upper halves.

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REFERENCES

- [1] M. M. Prieto, F. Fernández, and J. Rendueles, "Development of stepwise thermal model for annealing line heating furnace", *Ironmaking & steelmaking*, vol. 32, no 2, pp. 165-170, 2005. **(Journal Paper)**
- [2] N. Depree, J. Sneyd, S.Taylor, M. P. Taylor, J.J.J. Chen, S. Wang, M. OConner, "Development and validation of models for annealing furnace control from heat transfer fundamentals", *Computers & Chemical Engineering*, vol 34, no. 11, pp. 1849-1853, 2010. **(Journal Paper)**
- [3] S. Strommer, M. Niederer, A. Steinboeck, and A. Kugi. "A mathematical model of a direct-fired continuous strip annealing furnace", *International Journal of Heat and Mass Transfer*, vol. 69, pp. 375-389, 2014. **(Journal Paper)**
- [4] S. Zareba, A. Wolff, and M. Jelali, "Mathematical modelling and parameter identification of a stainless steel annealing furnace" *Simulation Modelling Practice and Theory*, vol. 60, pp. 15-39, 2016. **(Journal Paper)**
- [5] A. Jaklič, T. Kolenko, and B. Zupančič, "The influence of the space between the billets on the productivity of a continuous walking-beam furnace", *Applied Thermal Engineering*. vol. 25, pp. 783-795, 2005. **(Journal Paper)**
- [6] S.H. Han, and D. Chang, "Radiative slab heating analysis for various fuel gas compositions in an axial-fired reheating furnace", *International Journal of Heat and Mass Transfer*, vol. 55, pp. 4029-4036, 2012. **(Journal Paper)**
- [7] Singh, V.K. and P. Talukdar, "Comparisons of different heat transfer models of a walking beam type reheat furnace", *International Communications in Heat and Mass Transfer*, vol. 47, pp. 20-26, 2013. **(Journal Paper)**
- [8] T. Morgado, P.J. Coelho, and P. Talukdar, "Assessment of uniform temperature assumption in zoning on the numerical simulation of a walking beam reheating furnace", *Applied Thermal Engineering*. vol. 76, pp. 496-508, 2015. **(Journal Paper)**
- [9] S.A. Arreola-Villa, G. Solorio-Diaz, H. J. Vergara-Hernandez, O. Vazques-Gomez "Numerical simulation of a metallic load in continuous movement during heating in an experimental vertical-type furnace", *Journal of the Brazilian Society of Mechanical Sciences and Engineering*, vol. 40, pp. 280, 2018. **(Journal Paper)**